Yiwei ZHAO | 赵奕炜

Curriculum Vitae

Indiana University Bloomington
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Last modified in November 2026

RESEARCH INTERESTS

My research areas include and are not limited to: Parallel & Distributed Computing; Computer Systems & Architecture; High-Performance Computing; Databases; Algorithmic Theory; Machine Learning Systems.

PROFESSION EXPERIENCE

Indiana University Bloomington, Bloomington, Indiana, USA.

- Tenure-Track Assistant Professor, October 2026 – present
  Department of Engineering; Luddy School of Informatics, Computing, and Engineering.

Meta Platforms, Inc., Redmond, Washington, USA.

- Research Scientist Intern, May 2025 – August 2025, and May 2023 – November 2023

EDUCATION

Carnegie Mellon University, Pittsburgh, Pennsylvania, USA, 2021 – 2026.

- Ph.D. in Electrical and Computer Engineering.
- Advisor: Phillip B. Gibbons.
- Thesis: Theoretically- and Practically-Efficient Processing-in-Memory Algorithms and Systems.

Tsinghua University, Beijing, China, 2017 – 2021.

- Bachelor in Electronic Engineering. Graduated with highest honor.
- Double major in Economics & Finance.

HONORS & AWARDS

- Michel and Kathy Doreau Graduate Fellowship (2024 - 2025).
- Lee-Stanziale Ohana Fellowship (2023 - 2024).
- Qualcomm PhD Fellowship Finalist (2024).
- Best Paper Runner-up, VLDB (2023).
- Carnegie Institute of Technology Dean's Fellowship, (2021 - 2022).
- Outstanding Graduate, Tsinghua University & Beijing Municipality (2021).
- China National Scholarship for Undergraduates (2018 - 2020).

PUBLICATIONS

Full Publications

[1] Hongbo Kang, Xiangyun Ding, **Yiwei Zhao**, Yingdi Shan, Guy E. Blelloch, Laxman Dhulipala, Yan Gu, Charles McGuffey, Mingxing Zhang, Yongwei Wu, and Phillip B. Gibbons. 2027. "**FH-index: A Finger-based Learned Index**". In Proceedings of the VLDB Endowment (**PVLDB**), vol.20.

[2] Hyoungjoo Kim, **Yiwei Zhao**, Rutwik Pandit, Matteo Interlandi, Carlo Curino, Andrew Pavlo, and Phillip B. Gibbons. 2027. "**These Indexes Hit Hard: Toward Practical GPU Indexes with Concurrent Dynamic Updates and Variable-Length Key/Values**". Conditionally accepted by **SIGMOD** '27.

[3] Hongbo Kang, **Yiwei Zhao**, Kunal Agrawal, Yongwei Wu, and Phillip B. Gibbons. 2026. "**Non-Clairvoyant Scheduling for Processing-in-Memory**". In Proceedings of the 38th ACM Symposium on Parallelism in Algorithms and Architectures (**SPAA** '26). Association for Computing Machinery, New York, NY, USA, 392-404. doi:10.1145/3816782.3819200.

[4] **Yiwei Zhao**, Hongbo Kang, Ziyang Men, Yan Gu, Guy E. Blelloch, Laxman Dhulipala, Charles McGuffey, and Phillip B. Gibbons. 2026. "**PIM-zd-tree: A Fast Space-Partitioning Index Leveraging Processing-in-Memory**". In Proceedings of the 31st ACM SIGPLAN Annual Symposium on Principles and Practice of Parallel Programming (**PPoPP** '26). Association for Computing Machinery, New York, NY, USA, 480-495. doi:10.1145/3774934.3786411.

[5] **Yiwei Zhao**, Hongbo Kang, Yan Gu, Guy E. Blelloch, Laxman Dhulipala, Charles McGuffey, and Phillip B. Gibbons. 2025. "**Optimal Batch-Dynamic kd-trees for Processing-In-Memory with Applications**". In Proceedings of the 37th ACM Symposium on Parallelism in Algorithms and Architectures (**SPAA** '25). Association for Computing Machinery, New York, NY, USA, 350-366. doi:10.1145/3694906.3743318.

[6] Hyoungjoo Kim, **Yiwei Zhao**, Andrew Pavlo, and Phillip B. Gibbons. 2025. "**No Cap, This Memory Slaps: Breaking Through the Memory Wall of Transactional Database Systems with Processing-in-Memory**". In Proceedings of the VLDB Endowment (**PVLDB**), 18(11): 4241-4254, July 2025. doi:10.14778/3749646.3749690.

[7] **Yiwei Zhao**, Jinhui Chen, Sai Qian Zhang, Syed Shakib Sarwar, Kleber Hugo Stangherlin, Jorge Tomas Gomez, Jae Sun Seo, Barbara De Salvo, Chiao Liu, Phillip B. Gibbons, and Ziyun Li. 2025. "**H4H: Hybrid Convolution-Transformer Architecture Search for NPU-CIM Heterogeneous Systems for AR/VR Applications**". In Proceedings of the 30th Asia and South Pacific Design Automation Conference (**ASPDAC** '25). Association for Computing Machinery, New York, NY, USA, 1133-1141. doi:10.1145/3658617.3697627. arXiv:2410.08326.

- Extended Abstract: In Highlights of Parallel Computing Workshop (**HOPC** '25). doi:10.1145/3746238.3746241.
- Poster: In 61th ACM/IEEE Design Automation Conference (**DAC** '24), 2024.

[8] Hongbo Kang, **Yiwei Zhao**, Guy E. Blelloch, Laxman Dhulipala, Yan Gu, Charles McGuffey, and Phillip B. Gibbons. 2023. "**PIM-trie: A Skew-Resistant Trie for Processing-in-Memory**". In Proceedings of the 35th ACM Symposium on Parallelism in Algorithms and Architectures (**SPAA** '23). Association for Computing Machinery, New York, NY, USA, pp. 1-14. doi:10.1145/3558481.3591070.

[9] Hongbo Kang, **Yiwei Zhao**, Guy E. Blelloch, Laxman Dhulipala, Yan Gu, Charles McGuffey, and Phillip B. Gibbons. 2022. "**PIM-tree: A Skew-resistant Index for Processing-in-Memory**". In Proceedings of the VLDB Endowment (**PVLDB**), 16(4): 946-958, December 2022. doi:10.14778/3574245.3574275. arXiv:2211.10516. ***Best Research Paper Runner-up in VLDB 2023***.

- Journal Version: In the VLDB Journal (**VLDBJ**), 2025. doi:10.1007/s00778-025-00937-5.
- Extended Abstract: In Workshop on Highlights of Parallel Computing (**HOPC** '23). doi:10.1145/3597635.3598029.

[10] Zeyan Li, Junjie Chen, Yihao Chen, Chengyang Luo, **Yiwei Zhao**, Yongqian Sun, Kaixin Sui, Xiping Wang, Dapeng Liu, Xing Jin, Qi Wang, and Dan Pei. 2023. "**Generic and Robust Root Cause Localization for Multi-Dimensional Data in Online Service Systems**". In Journal of Systems and Software (**JSS**), Vol. 203, (2023), 111748. doi:10.1016/j.jss.2023.111748.

[11] Zeyan Li, Chengyang Luo, **Yiwei Zhao**, Yongqian Sun, Kaixin Sui, Xiping Wang, Dapeng Liu, Xing Jin, Qi Wang, and Dan Pei. 2019. "**Generic and Robust Localization of Multi-Dimensional Root Cause**". In the 30th International Symposium on Software Reliability Engineering (**ISSRE** '19). Oct. 28-31, 2019, Berlin. doi:10.1109/ISSRE.2019.00015.

Preprints & Works Under Review

[12] **Yiwei Zhao***, Qiushi Lin*, Hongbo Kang, Guy E. Blelloch, Laxman Dhulipala, Yan Gu, Charles McGuffey, and Phillip B. Gibbons. "**TD-Orch: Efficient Task-Data Orchestration for Distributed Systems with Application to Graph Processing**". arXiv:2511.11843.

[13] **Yiwei Zhao**, Yi Zheng, Huapeng Su, Jieyu Lin, Stefano Ambrogio, Cijo Jose, Michael Ramamonjisoa, Patrick Labatut, Barbara De Salvo, Chiao Liu, Phillip B. Gibbons, and Ziyun Li. "**AdaDINO: Context-Adaptive DINO-Distilled Vision Foundation Models for Efficient Open-Vocabulary Edge Inference**". arXiv:2604.15622.

[14] Qiushi Lin, Phillip B. Gibbons, Jovan Stojkovic, and **Yiwei Zhao**. "**Gutenberg: Taming Latency-Critical Cloud Services with Near-Data-Processing**". arXiv:2609.06691.

[15] Mohammad Bakhshalipour, Valerie Choung, Rutwik Pandit, Sean Hyacinthe, **Yiwei Zhao**, and Phillip B. Gibbons. "**Benchmarking Memory Systems with a Diverse Zoo: Where Does Time Go?**".

SERVICES

- **Program Committee**: ISCA'27, SPAA'27.
- **External/Reciprocal Program Committee**: VLDB'27, SIGMETRICS'27, SPAA'26, PACT'26, ESA'26, PACT'25, SPAA'25.
- **Artifact Evaluation Committee**: ALENEX'26.
- **Student Council for Departmental Faculty Hiring**: Chair, January 2024 – June 2025, CMU.
- **Student Council for Departmental Faculty Hiring**: Member, January 2023 – January 2024, CMU.

INVITED TALKS

(All conference paper presentations are excluded due to space limit; please refer to the *Publications* for the complete list.)

- **October 2025, UC Berkeley, Simons Institute for the Theory of Computing. In Algorithmic Foundations for Emerging Computing Technologies**: Optimal Semi-Balanced Trees for Processing-in-Memory.
- **October 2025. UC Berkeley. In Programming System Seminar**: Building HPC Systems for Near-Data-Processing: Theory and Practice.
- **October 2025, Stanford University. In Software Research Seminar**: Building Programming Systems for Near-Data-Processing: Theory and Practice.
- **October 2024, CMU. In Parallel Data Lab (PDL) Retreat**: Fast and Principled Techniques for Heterogeneous Compute and Memory.
- **November 2023, CMU**. **In Parallel Data Lab (PDL) Retreat**: System Design on Processing-In-Memory: Starting from Database Systems.
- **November 2022, CMU**. **In Parallel Data Lab (PDL) Retreat**: PIM-tree: A Theoretically and Practically Efficient Index for Processing-In-Memory.

TEACHING EXPERIENCES

- **18-751 Applied Stochastic Processes, with Applications to AI/ML**: Teaching Assistant, Fall 2024, CMU.
- **18-742 Computer Architecture and Systems**: Teaching Assistant, Spring 2024, CMU.

REFERENCES

Phillip B. Gibbons (Ph.D. Advisor)
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